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Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Joncryl® U 4501**

(ID no. 30736819/SDS_GEN_DE/EN)

Date of print 11.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Joncryl® U 4501

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Binder

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Business Unit Resins and Additives

Telephone: +49 621 60-0
E-mail address: ed-psr@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

No need for classification according to GHS criteria for this product.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

The product does not require a hazard warning label in accordance with GHS criteria.

According to Regulation (EC) No 1272/2008 [CLP]

Labeling of special preparations (GHS):

EUH208: May produce an allergic reaction. Contains: Adipohydrazide, 3-aminomethyl-3,5,5-trimethylcyclohexylamine, reaction mass of: 5-chloro-2- methyl-4-isothiazolin-3-one [EC no. 247-500-7]and 2-methyl-2H -isothiazol-3- one [EC no. 220-239-6] (3 :1)

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

No specific dangers known, if the regulations/notes for storage and handling are considered. Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605. The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

Aqueous dispersion of a polymer based on:polyurethane

Regulatory relevant ingredients

reaction mass of: 5-chloro-2- methyl-4-isothiazolin-3-one [EC no. 247-500-7]and 2-methyl-2H - isothiazol-3- one [EC no. 220-239-6] (3 :1)

Content (W/W): >= 0 % - < 0,0015 % CAS Number: 55965-84-9 REACH registration number: 01-2120764691-48 INDEX-Number: 613-167-00-5	Acute Tox. 3 (oral) Acute Tox. 2 (Inhalation - mist) Acute Tox. 2 (dermal) Skin Corr./Irrit. 1C Eye Dam./Irrit. 1 Skin Sens. 1A Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 100 M-factor chronic: 100 H301, H317, H314, H310 + H330, H400, H410 EUH071 Specific concentration limit: Skin Sens. 1A: >= 0,0015 % Eye Dam./Irrit. 1: >= 0,6 % Eye Dam./Irrit. 2: 0,06 - < 0,6 % Skin Corr./Irrit. 1C: >= 0,6 % Skin Corr./Irrit. 2: 0,06 - < 0,6 %

Adipohydrazide

Content (W/W): >= 0,3 % - < 1 % CAS Number: 1071-93-8 EC-Number: 213-999-5 REACH registration number: 01-2119962900-36	Aquatic Chronic 2 Skin Sens. 1 H317, H411

2-diethylaminoethanol

Content (W/W): >= 0,3 % - < 1 % CAS Number: 100-37-8 EC-Number: 202-845-2 REACH registration number: 01-2119488937-14 INDEX-Number: 603-048-00-6	Flam. Liq. 3 Acute Tox. 3 (Inhalation - vapour) Acute Tox. 4 (oral) Acute Tox. 3 (dermal) Skin Corr./Irrit. 1B Eye Dam./Irrit. 1 STOT SE 3 (irr. to respiratory syst.) H226, H302, H335, H314, H311 + H331 Specific concentration limit: STOT SE 3, irr. to respiratory syst.: >= 5 %

3-aminomethyl-3,5,5-trimethylcyclohexylamine

Content (W/W): < 0,001 % CAS Number: 2855-13-2 EC-Number: 220-666-8 REACH registration number: 01-2119514687-32 INDEX-Number: 612-067-00-9	Acute Tox. 4 (oral) Aquatic Chronic 3 Skin Corr./Irrit. 1B Eye Dam./Irrit. 1 Skin Sens. 1A H302, H317, H314, H412 Specific concentration limit: Skin Sens. 1A: >= 0,001 % Acute toxicity estimate: oral: 1.030 mg/kg

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
If difficulties occur after vapour/aerosol has been inhaled, remove to fresh air and seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting unless told to by a poison control center or doctor.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam

5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours
Advice: Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Further information:
Contaminated extinguishing water must be disposed of in accordance with official regulations.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing.

6.2. Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

No special measures necessary provided product is used correctly.

Protection against fire and explosion:
No special precautions necessary.

7.2. Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage class according to TRGS 510 (originally VCI, Germany): (12) Non-combustible liquids

Protect from temperatures below:5 °C
Protect from temperatures above:60 °C

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

100-37-8: 2-diethylaminoethanol
Short Term Exposure Classification: (TRGS 900 (DE))
Category I: Substances for which the localized effect has an assigned exposure limit or for substances with a sensitizing effect in respiratory passages
OEL 24 mg/m3 ; 5 ppm (TRGS 900 (DE))
Ceiling limit value/factor: 1
If the occupational exposure limit value (AGW) and the biological limit value (BGW) are complied with, there should be no risk of damage for the unborn child (see TRGS 900, Number 2.7)
Skin Designation (TRGS 900 (DE))
The substance can be absorbed through the skin.

Components with PNEC

2855-13-2: 3-aminomethyl-3,5,5-trimethylcyclohexylamine
freshwater: 0,06 mg/l
marine water: 0,006 mg/l
intermittent release: 0,23 mg/l
sediment (freshwater): 5,784 mg/kg
sediment (marine water): 0,578 mg/kg
soil: 1,121 mg/kg
STP: 3,18 mg/l
oral (secondary poisoning):
No PNEC oral derived, as accumulation in organisms is not to be expected.

100-37-8: 2-diethylaminoethanol
freshwater: 0,0623 mg/l
marine water: 0,00623 mg/l

intermittent release: 0,34 mg/l
sediment (freshwater): 0,673 mg/kg
sediment (marine water): 0,0673 mg/kg
soil: 0,0977 mg/kg
STP: 10 mg/l
oral (secondary poisoning):
According to EU risk assessment risks are negligible

1071-93-8: Adipohydrazide
No PNEC value available.

55965-84-9: reaction mass of: 5-chloro-2- methyl-4-isothiazolin-3-one [EC no. 247-500-7]and 2-methyl-2H -isothiazol-3- one [EC no. 220-239-6] (3 :1)
freshwater: 0,00339 mg/l
marine water: 0,00339 mg/l
STP: 0,23 mg/l
sediment (freshwater): 0,027 mg/kg
sediment (marine water): 0,027 mg/kg
soil: 0,01 mg/kg

Components with DNEL

2855-13-2: 3-aminomethyl-3,5,5-trimethylcyclohexylamine
worker: Long-term exposure - local effects, Inhalation: 0,073 mg/m3

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection not required.

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	semi translucent	
Odour:	faint odour, amine-like	
Odour threshold:	No applicable information available.	
Freezing point:	0 °C Information applies to the solvent.	
Boiling point:	100 °C Information applies to the solvent.	
Flammability:	not flammable	
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	No flash point - Measurement made up to the boiling point.	
Auto-ignition temperature:	Based on the water content the product does not ignite.	
Thermal decomposition:	Stable up to boiling point.	
SADT:	> 75 °C	
pH value:	approx. 8 - 8,5	(DIN ISO 976)
Viscosity, dynamic:	approx. 700 mPa.s (23 °C, 100 1/s)	(DIN EN ISO 2555)
Thixotropy:	not thixotropic	
Solubility in water:	readily soluble	
Solubility (qualitative) solvent(s):	organic solvents soluble	
Partitioning coefficient n-octanol/water (log Kow):	Study scientifically not justified.	
Vapour pressure:	23,4 hPa (20 °C) Information applies to the solvent.	
Density:	1,04 g/cm3 (20 °C)	(ISO 2811-3)
Relative vapour density (air):	not determined	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.

Other safety characteristics

Radioactivity: not radioactive for transport purposes

Hygroscopy: Non-hygroscopic

Surface tension: not determined

SAPT-Temperature: Product does not fulfil criteria for polymerizing substances according to transport regulations.

Evaporation rate: not determined

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

10.2. Chemical stability

The product is chemically stable.

Peroxides: The product does not contain peroxides. The product/the substance has not a tendency towards the formation of peroxide.

10.3. Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

10.4. Conditions to avoid

See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:
No substances known that should be avoided.

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Experimental/calculated data:
LD50 rat (oral): > 5.000 mg/kg
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LC50 rat (by inhalation): 4 h
not determined

LD50 rat (dermal):
not determined

Irritation

Experimental/calculated data:
Skin corrosion/irritation
rabbit: non-irritant (BASF-Test)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Serious eye damage/irritation
rabbit: non-irritant (BASF-Test)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Respiratory/Skin sensitization

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)
The product has not been tested. The statement has been derived from the properties of the individual components.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on the ingredients, there is no suspicion of a mutagenic effect.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Repeated oral uptake of the substance did not cause substance-related effects.

Repeated inhalative uptake of the substance did not cause substance-related effects.

Repeated dermal uptake of the substance did not cause substance-related effects.

The product has not been tested. The statement has been derived from the properties of the individual components.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 12: Ecological Information

12.1. Toxicity

Toxicity to fish:
LC50 (96 h) > 100 mg/l, Fish
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna (Screening (style of OECD 202), static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants:
EC50 (72 h), algae
No data available.

Microorganisms/Effect on activated sludge:
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Chronic toxicity to fish:
No data available regarding toxicity to fish.

Chronic toxicity to aquatic invertebrates:
No data available regarding toxicity to daphnids.

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
The polymer component of the product is poorly biodegradable.

12.3. Bioaccumulative potential

Bioaccumulation potential:
At the present state of knowledge, no negative ecological effects are expected.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface. The product has not been tested. The statement has been derived from the properties of the individual components.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

12.7. Other adverse effects

The product does not contain substances that are listed in Annex I of Regulation (EC) 2037/2000 on substances that deplete the ozone layer.

12.8. Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
According to experience, the material has no harmful effect on the environment.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number

See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Hazardous Incident Ordinance (Germany):
Listed in above regulation: no

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

Water hazard class (§8/§10 AwSV (Self-classification of the mixture according to calculation method)): (1) Weakly water polluting.

Law on the Protection of Working Youth
German Regulation TA Luft (Technical Instruction on Air Quality Control, i.e. first Directive to the Federal Immission Control Ordinance)

Annex VIII to Regulation (EC) No 1272/2008 - Poison Center Notifications: This product is not subject to submission obligations.
A voluntary submission is not planned.

15.2. Chemical Safety Assessment

Chemical Safety Assessment not required

SECTION 16: Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Flam. Liq.	Flammable liquids
STOT SE	Specific target organ toxicity — single exposure
H301	Toxic if swallowed.
H317	May cause an allergic skin reaction.
H314	Causes severe skin burns and eye damage.
H310 + H330	Fatal in contact with skin or if inhaled
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H226	Flammable liquid and vapour.
H302	Harmful if swallowed.
H335	May cause respiratory irritation.
H311 + H331	Toxic in contact with skin or if inhaled
H412	Harmful to aquatic life with long lasting effects.
EUH071	Corrosive to the respiratory tract.

Abbreviations

ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.